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October 26, 2022

Amherst Pelham Regional School District
Business Office
170 Chestnut Street
Amherst, MA 01002

Attn: Mr. Rupert Roy-Clark
C: (413) 362-1855
E: roy-clarkr@arps.org

Re: APRMS Auditorium Framing Analysis for Photovoltaic Panels
Amherst Pelham Regional Middle School
170 Chestnut Street, Amherst, MA
Gale JN 839820

Dear Mr. Roy-Clark:

In accordance with our proposal, Gale Associates, Inc. (Gale) has performed a visual and limited hands-on evaluation and structural analysis of the auditorium steel roof framing to support photovoltaic (PV) panels at the above-referenced facility. To conduct this evaluation, representatives from Gale visited the project site on June 1, 2022. Access to the facility and ladder access to the roof framing components was provided by Amherst Pelham Regional Middle School (APRMS) staff.

The purpose of this evaluation was to observe the existing conditions of the roof framing members and associated components, perform local field measurements to cross-reference the structural roof framing plans supplied by APRPS, and to provide an opinion on the feasibility of installing PV panels on the auditorium roof. This letter provides Amherst Pelham Regional School District (APRSD) with a summary of our findings.

Background Information

The Amherst Pelham Middle School was reportedly constructed in the late 1960s and officially opened in 1969. It is our understanding the most recent roof renovation was completed in 2002. The Middle School is comprised of three (3) buildings, connected by corridor spaces. The central structure, the auditorium, and surrounding areas were included our structural evaluation. The existing auditorium roof covering is a thermoplastic roof membrane system, which is supported on a 4-1/2-inch deep metal roof deck. The primary roof framing members of the auditorium are comprised of steel bar joists and wide-flange beams spanning between a combination of steel wide-flange girders and two (2) central steel trusses. APRMS provided Gale with original architectural and structural drawings prepared by Alderman & MacNeish, dated August 14, 1967, to cross-reference with the representative measurements performed in the field and to utilize for Gale's structural analysis of the roof framing to support the proposed PV panels.

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Observations

Gale made the following general observations during the visual evaluation of accessible roof framing components:

Auditorium

1. The roof framing observed in accessible areas appears to consist of a steel beam, bar joist, and truss framing system. Main framing members span the width of the auditorium in the north-south direction, bearing on steel columns at each end. The main framing members were observed to be either 27-inch deep steel beams, or trusses type "T-1" or "T-2" as designated by the original construction drawings by Alderman & MacNeish. Secondary framing members were observed to be 14-inch deep steel beams spaced approximately 5 foot, 3 inches on center, and 12- and 16-inch deep joists spaced at approximately 4 foot, zero inches on center, spanning between the main framing members in the east-west direction. Horizontal top and bottom chord steel bridging was observed between the steel joists throughout. The existing structural drawings indicate a 4-1/2-inch deep metal roof deck, which appeared to be consistent with the destructive roof test cuts performed by Gale in a previous evaluation. Suspended elements including, but not limited to, lighting, duct work, fire suppression, various auditorium equipment, and acoustic gypsum ceiling, were observed to be hung directly to the underside of the secondary framing members. Gale representatives observed that the framing of the auditorium roof area appeared to be generally consistent with what is shown on the existing structural drawings, dated August 14, 1967.



Photo 1: Overall view of auditorium seating with acoustic ceiling.



Photo 2: Overall view of Auditorium stage.



Photo 3: Overall view of 14-inch deep steel roof beams located above the Auditorium stage.



Photo 4: Overall view of equipment and utilities suspended directly from the underside of 14-inch deep steel beams located above the Auditorium stage, looking north along Column Line E.

2. In general, the accessible steel roof framing observed at the time of the inspection appeared to be in good condition without areas of extensive damage or severe corrosion. Isolated areas of efflorescence staining were observed in some locations, as depicted in Photo 5, indicating signs of potential water infiltration. The extent of observed efflorescence appears to be localized and does not appear to have compromised the structural integrity of the observed members.



Photo 5: View of efflorescence staining on 14-inch deep steel beam located above Auditorium stage, looking south along Column Line E.



Photo 6: View of 14-inch deep steel beams in good condition.

Band and Chorus Rooms

1. The roof framing system observed in accessible areas above the Band and Chorus Rooms, appears to consist of 4-1/2-inch deep metal roof deck supported by 16-inch deep steel joists spaced at approximately 4 foot, zero inches on-center. Steel joists were observed to span in the east-west direction between a combination of 14-inch deep steel beams of various span lengths. The 14-inch deep steel beams span in the north-south direction between wide-flange steel columns. Columns were not observed in the field and the sizes were assumed based on the drawings provided. Gale representatives observed that the roof framing of the Band and Chorus areas appeared to be generally consistent with the information shown on the existing structural drawings, dated August 14, 1967.



Photo 7: Overall view of 16-inch deep steel joists with top chord horizontal bridging located above the Chorus room, looking east along Column Line 39.



Photo 8: View of typical 16-inch deep steel joists and 14-inch deep steel beam beyond.



Photo 9: Typical view of a steel joist connection to a 14-inch deep steel beam, looking north along Column Line E.



Photo 10: View of ceiling tiles and utilities suspended from steel joists, located above the Band room.

2. In general, the steel roof framing observed in the accessible locations at the time of the inspection at the east end of the building appeared to be in good condition. Isolated areas of minor corrosion were observed above the Band Room as shown in Photo 11. Miscellaneous breaches in the metal roof deck were observed above the Chorus Room as shown in Photo 12.



Photo 11: View of minor corrosion at a steel joist located above the Chorus Room.



Photo 12: Typical view of scattered breaches in the metal roof deck above Band Room.

Mechanical Room/ Crawl Space

1. The roof framing system observed in accessible areas in the mechanical room and adjacent crawl space appears to consist of 4-1/2-inch deep metal roof deck supported by 14-inch deep steel joists spaced at approximately 4 foot, zero inches on-center. Steel joists appear to span in the east-west direction between main framing members consisting of 14-inch deep steel girders. The 14-inch deep steel beams span in the north-south direction between wide-flange columns. Gale observed top and bottom chord horizontal bridging spaced at approximately 6 foot, zero inches on center, spanning north-to-south. Gale observed large deck penetrations above the top chord of multiple joists, exposing the underside of a steel beam welded to each side of the joist. Refer to Photo 16. This observation appears consistent with the 1994 roofing renovation drawings provided to Gale, designed by Tessier Associates, Inc., dated May 17, 1994. Original framing observed at this roof area appears to be generally consistent with what is shown on the existing structural drawings, dated August 14, 1967.



Photo 13: Overall view of mechanical room and crawl space entrance.



Photo 14: View of accessible steel roof framing in the mechanical room.



Photo 15: Overall view of crawl space.

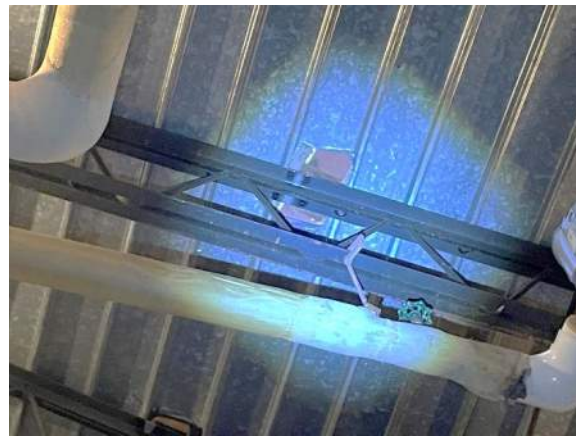


Photo 16: View of deck penetration with beam welded above.

2. In general, the observed, accessible steel roof framing at the east end of the building appeared to be in good condition with isolated areas of minor corrosion. Isolated areas of cracking were observed at the concrete masonry wall (CMU) wall beam pocket as shown in Photos 17 and 18. These areas were not included in Gale's roof framing evaluation and should be monitored by APRSD.

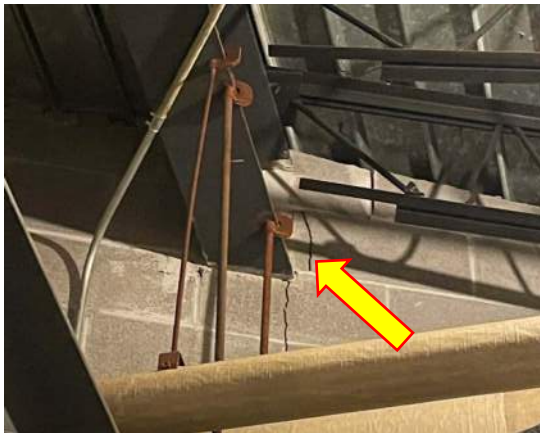


Photo 17: View of a crack in the CMU wall at the south bearing end of a 14-inch deep steel beam.

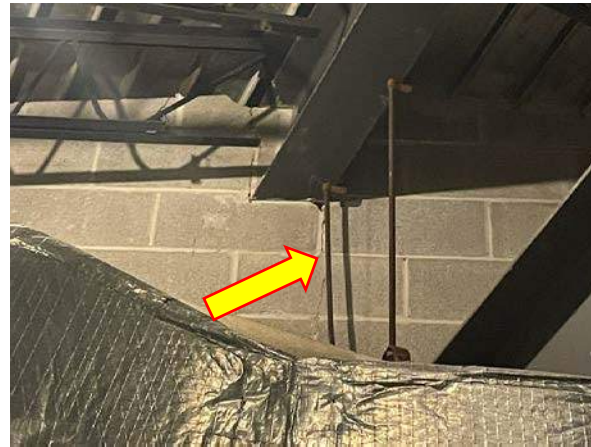


Photo 18: View of a crack in the CMU wall at the north bearing end of a 14-inch deep steel beam.

Discussion & Opinions

The observed steel framing was generally in good condition and the field measurements performed by Gale of representative areas were found to be generally consistent with the existing structural drawings, dated August 4, 1967, and the roofing renovation drawings, dated May 17, 1994. A limited structural analysis was performed utilizing the above-referenced drawings. The Massachusetts State Building Code (MSBC) 9th Edition is applicable for this project. The MSBC 9th Edition adopts the International Existing Building Code (IEBC) 2015 with select MSBC amendments. The IEBC requires a structural analysis to be performed on roof framing members where the gravity load of the member is increased by 5% or more by any type of alteration. Gale calculated that the proposed roof renovation system and addition of a PV array would increase the loading on roof framing members above the Auditorium, Band, and Chorus rooms by more than the 5% threshold specified by the IEBC. The following are the results of the limited structural analysis performed by Gale, and our opinion of the structural capacity of the existing framing members to support the PV array and proposed roof system replacement.

Gale's analysis considered estimated dead loads including the proposed roof system replacement, suspended loads of equipment and utilities, the self-weight of the supporting structural element, code-prescribed snow loads, and an assumed 4 pounds per square foot (psf) for the weight of a mechanically attached PV array. Gale's snow load analysis determined that the required sloped roof snow load is 35 psf in accordance with the MSBC 9th Edition Amendments, which govern over the snow loads calculated in accordance with the IEBC 2015 requirements. The design roof live load included on Sheet S-1 of the existing structural drawings, dated April 4, 1967, is 30 psf. Including the proposed snow load defined by the MSBC 9th Edition would increase the snow load applied to each existing member by 5 psf. The allowable total loading as stated on Sheet S-1 for the deck and joists are 52 psf and 60 psf, respectively. Gale observed that all interior suspended equipment and utilities were attached directly to the underside of the joists. To account for this loading, Gale used the estimated loading criteria shown in Table 1.

SUSPENDED ELEMENT	ESTIMATED HUNG LOAD
Acoustic Drop Ceiling Panels	3 psf
Sprinkler Piping (fully charged)	3 psf
Ductwork	1 psf
Lighting	1 psf
Auditorium Misc. Equipment	2 psf
TOTAL	10 psf

Table 1: Suspended Loading of Auditorium Equipment & Utilities

Based on Gale's limited structural analysis, it is Gale's opinion that some of the existing steel joists of the auditorium framing do not have adequate capacity to support the addition of a PV array. Figure 1 below shows the location of the joists Gale determined to be overstressed by the proposed additional loading of the PV array, without augmentation. Figure 2 below shows the roof locations where it is Gale's opinion that the existing roof framing can adequately support the proposed loading of the PV array, without augmentation.

Furthermore, the existence of efflorescence below the roof deck indicates past water infiltration originating from the roof level. It is Gale's opinion that the roof system be repaired, partially replaced, or fully replaced to reduce future water infiltration and to extend the service life of the structure.

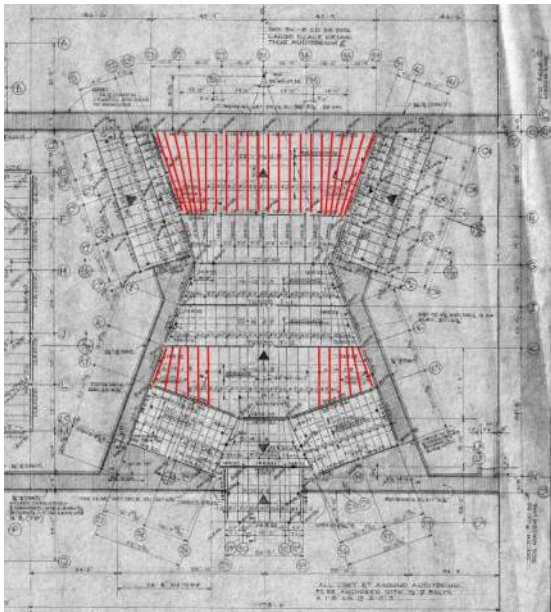


Figure 1: Framing plan with overstressed joists highlighted in red



Figure 2: Aerial view with allowable PV Array areas in green



If APRSD desires to install PV array on areas deemed to have inadequate structural capacity to support the PV array by Gale's limited structural analysis, augmentation options would likely include installing supplemental joists or beams between the existing framing members or reinforcing the existing joists by welding supplemental steel plates for increased strength. A benefit of reinforcing the existing joists would be that work could be performed without roof deck removal; however, it would be labor intensive and may not be cost efficient. A benefit of installing supplemental beams and joists would be that it may be less labor intensive, but it may require removal and replacement of the existing roof deck. Design of the structural augmentation was not included in Gale's scope of services for this phase of the project. At the request of APRSD, Gale can provide a proposal for design budget construction cost estimates of structural framing augmentation to support PV arrays in the areas deemed have inadequate capacity noted in Figure 1. Note that this analysis was limited, and further review of the framing may be necessary to develop a structural augmentation design.

It should be noted that it is Gale's opinion that the main framing members, including trusses, girders, and columns, have sufficient reserve capacity to support the additional loading of the PV array and proposed roof system replacement.

We trust this information suits your needs at this time. Please do not hesitate to contact us if you require additional information regarding this matter.

Best regards,

GALE ASSOCIATES, INC.

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